

Task – 2: JavaScript Mini Project(Calculator)

Introduction

This project is a simple Calculator Application developed using HTML5, CSS3, and JavaScript. The calculator performs basic arithmetic operations such as addition, subtraction, multiplication, and division through an interactive user interface. The application uses JavaScript DOM manipulation to handle user inputs and perform calculations dynamically.

Objective

The objective of this project is to understand JavaScript fundamentals, DOM manipulation, event handling, and interactive web application development by building a functional calculator.

Technologies Used

- HTML5
- CSS3
- JavaScript

Features

- User-friendly calculator interface
- Addition operation
- Subtraction operation
- Multiplication operation
- Division operation
- Decimal number support
- Clear (C) button
- Backspace (⌫) functionality
- Error handling for invalid expressions
- Responsive design

Working Process

User Input

Users can click number and operator buttons to create mathematical expressions.

DOM Manipulation

JavaScript updates the calculator display dynamically using DOM methods.

Calculation

When the "=" button is pressed, JavaScript evaluates the entered expression and displays the result.

Error Handling

Invalid expressions are detected and an alert message is displayed.

Sample Code

```
function appendValue(value){  
    document.getElementById("display").value += value;  
}
```

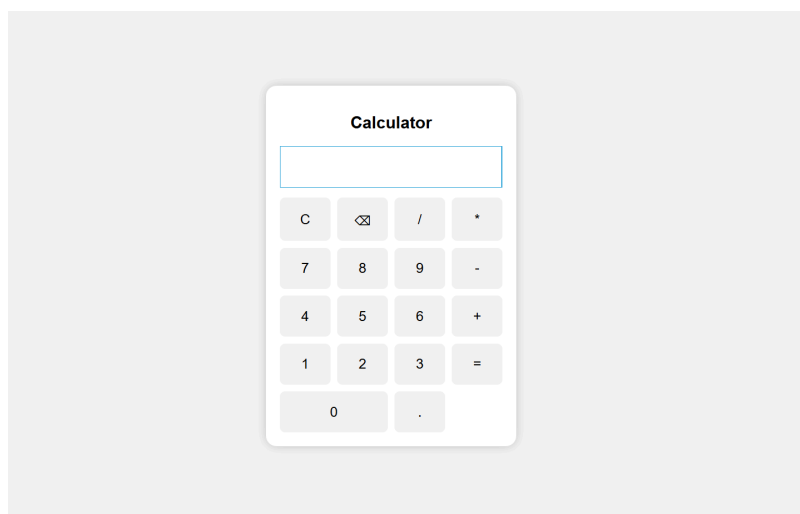
```
function clearDisplay(){  
    document.getElementById("display").value = "";  
}
```

```
function backspace(){  
    let display = document.getElementById("display");  
    display.value = display.value.slice(0, -1);  
}
```

```
function calculate(){  
    try{  
        document.getElementById("display").value =  
            eval(document.getElementById("display").value);  
    }  
    catch{  
        alert("Invalid Expression");  
    }  
}
```

Screenshots

Calculator Interface



GitHub Repository Link

GitHub Repository:

<https://github.com/charusmirtha/-InternSpark-Internship/tree/main/calculator-app>

Hosted Project Link

Live Website: <https://charusmirtha.github.io/-InternSpark-Internship/calculator-app>

Conclusion

This project provided practical experience in JavaScript programming and DOM manipulation. By developing a calculator application, I gained a better understanding of event-driven programming, user interaction, and responsive web development. The project strengthened my problem-solving skills and confidence in creating functional web applications.